

PROMPT ENGINEERING ASSINGMENT

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AI's used:

For managing PDF: DeepSeek

For trying out prompts: ChatGPT

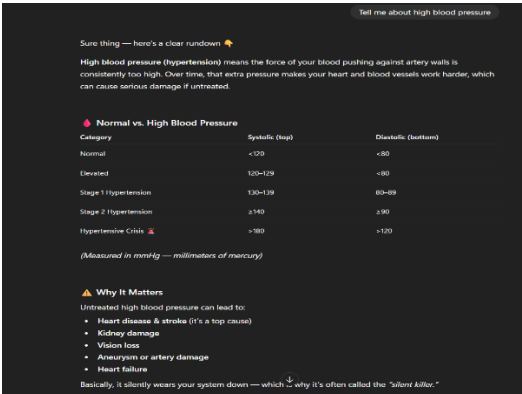
Part 1: Prompt Comparison Analysis

Chosen Domain : Healthcare.

Task : Draft a patient education handout for adults newly diagnosed with hypertension.

Version A

Vague Prompt : Tell me about high blood pressure.



Version B

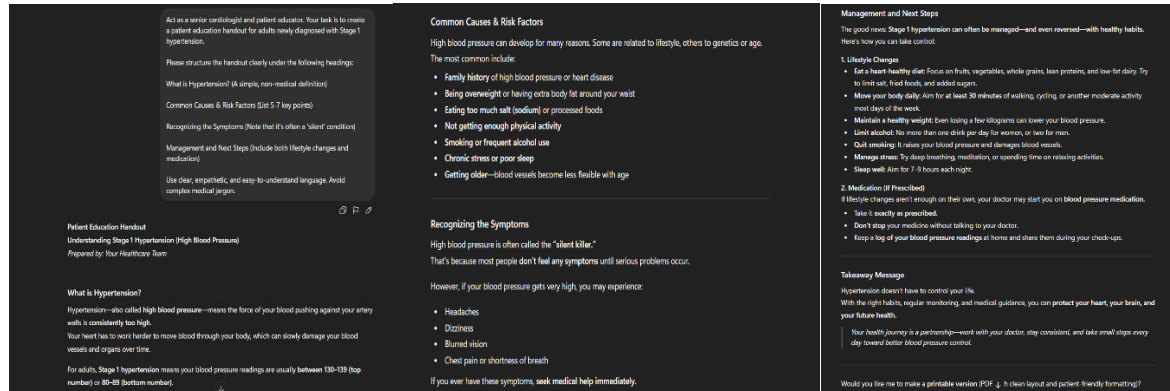
Good Prompt : Act as a senior cardiologist and patient educator. Your task is to create a patient education handout for adults newly diagnosed with Stage 1 hypertension.

Please structure the handout clearly under the following headings:

1. **What is Hypertension?** (A simple, non-medical definition)
2. **Common Causes & Risk Factors** (List 5-7 key points)

3. **Recognizing the Symptoms** (Note that it's often a 'silent' condition)
4. **Management and Next Steps** (Include both lifestyle changes and medication)

Use clear, empathetic, and easy-to-understand language. Avoid complex medical jargon.



I used **[ROLE PROMPTING]** This tells the AI to adopt a specific expertise and perspective, **[STRUCTURED OUTPUT]**, This forces the AI to organize the information in a logical, easy-to-follow format and **[DELIMITERS & CONSTRAINTS]** These instructions set clear boundaries and rules for the AI's response.

Comparison Analysis

The two answers from the AI are very different.

The first answer from the simple prompt (Version A) is not very good. It is a short text with many difficult words. It is not easy to read. A new patient will not understand it well. It does not help the patient know what to do.

The second answer from the better prompt (Version B) is very good. It is a clear and helpful text for a patient. I see three good things in Version B:

1. **It is organized.** It has titles like "What is Hypertension?" This helps you find information fast.
2. **It uses easy words.** It explains everything simply. A normal person can understand it.
3. **It tells you what to do.** It gives a list of actions, like "eat less salt." It is very practical.

Why was Version B better? Because the prompt was better.

I told the AI to "Act as a doctor." This is **Role Prompting**. So, the AI wrote like a kind doctor talking to a patient.

I also said "use these titles." This is **Structured Output**. So, the AI made a well-organized text with sections.

Finally, I said "use easy language." These were the **Rules**. So, the AI did not use difficult words.

A good prompt gives the AI a clear job. This helps the AI give a much better answer.

Part 2: Practical Prompt Design

Framework Selection & Justification

The CO-STAR framework was very useful. It helped me organize my thoughts and create a better prompt. It made me remember the target audience and the right tone for the posts.

I would use CO-STAR again for a social media plan. It is a good fit for this job. I don't need another framework because this one has all the important parts.

I learned that a framework acts like a guide for the AI. It tells the AI exactly what you need. The key lesson is that being clear about your audience and style gives you a much better result.

Prompt:

CONTEXT:

I have a small coffee shop. I need a social media plan for Instagram for the next month.

OBJECTIVE:

Make a calendar for 4 weeks. Give me 3 post ideas for each week. The goal is to get more customers to like my posts and visit my shop.

STYLE:

Write like a friendly person. Use some emojis. 😊

TONE:

Be nice and warm, not like a formal advertisement.

AUDIENCE:

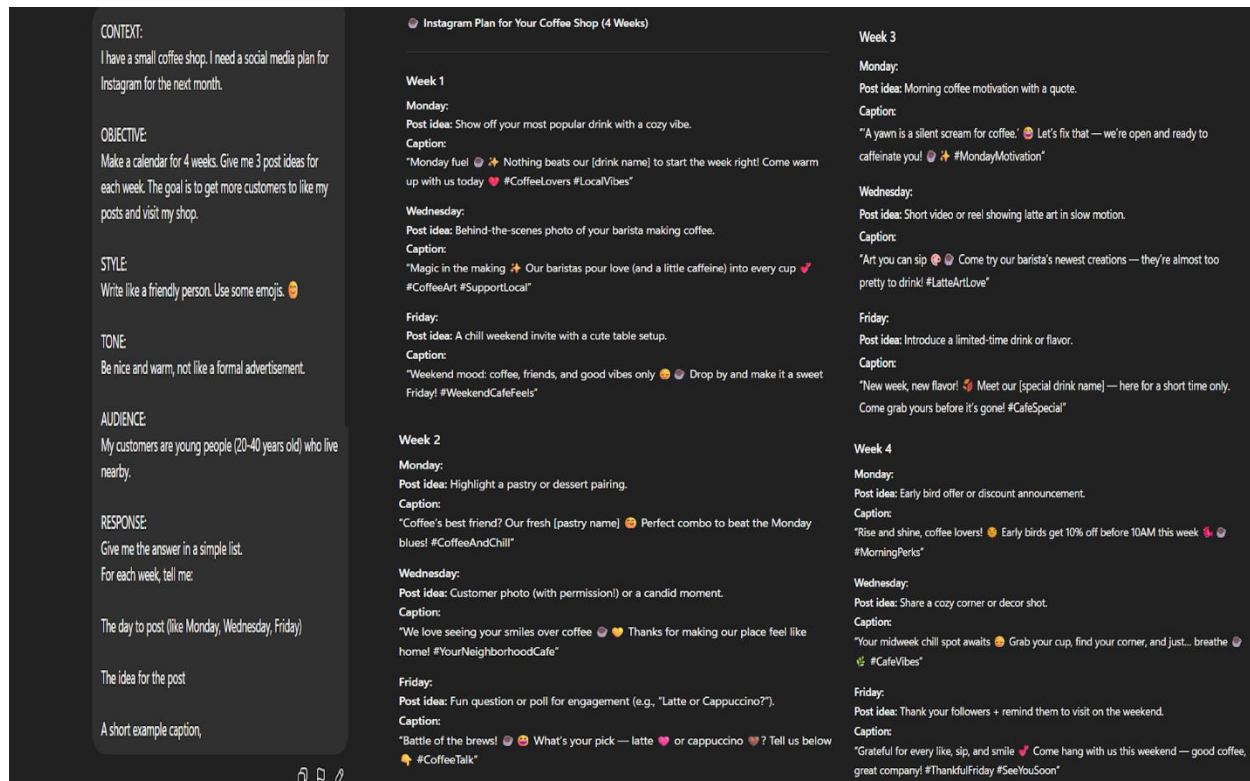
My customers are young people (20-40 years old) who live nearby.

RESPONSE:

Give me the answer in a simple list.

For each week, tell me:

- The idea for the post
- A short example caption



The AI successfully followed the framework. For example:

Audience & Tone: The captions use friendly and simple words, like "Good morning, everyone!" and "We love our community! ❤️". This matches the **casual tone** and is perfect for a **young local audience**.

Style: The AI used emojis like ☕ and 👍 in the captions, just like I asked in the **Style** part.

Response Format: The AI gave me the calendar in a **simple list** with the days of the week and post ideas, which is exactly what I asked for in the **Response** section.

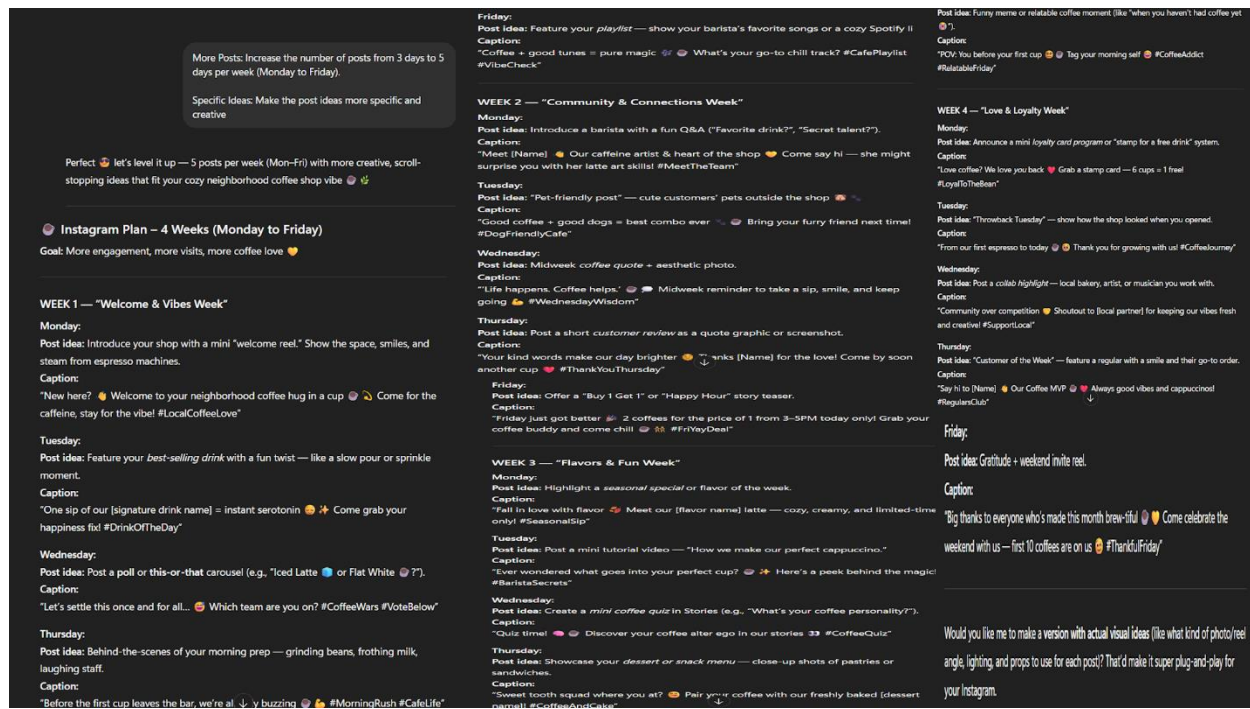
Weaknesses in the Initial Output:

There are only 3 post days per week, which is not enough.

The post ideas are too general and not creative.

Refined Prompt:

Thanks for the first calendar! Now, please improve it with these changes: **More Posts:** Increase the number of posts from 3 days to 5 days per week (Monday to Friday). **Specific Ideas:** Make the post ideas more specific and creative. For example, instead of "post a coffee photo," suggest ideas like "a behind-the-scenes video of making latte art" or "a customer photo contest."



After the new prompt, two main things changed:

- What changed:** The calendar now has **5 posts per week** (Monday to Friday).
Why it's better: This is better because posting more often keeps people interested in the coffee shop.
- What changed:** The post ideas are now **more specific and creative** (for example, "Meet the Barista" or "Coffee Trivia Tuesday").
Why it's better: This is better because it gives me clear and fun ideas. It is easier for me to know what to post, and people will like it more."

Framework Effectiveness Reflection

The CO-STAR framework was very useful. It helped me organize my thoughts and create a better prompt. It made me remember the target audience and the right tone for the posts. I would use CO-STAR again for a social media plan. It is a good fit for this job. I don't need another framework because this one has all the important parts. I learned that a framework acts like a guide for the AI. It tells the AI exactly what you need. The key lesson is that being clear about your audience and style gives you a much better result.

PART 3: CRITICAL REFLECTION

a) Skill Progression

At the start, I asked the AI simple questions. Now I know I must give clear instructions. My big "aha moment" was in Part 1. I saw that a better prompt gives a much better answer from the same AI. The hardest part was learning the frameworks like CO-STAR. I solved this by practicing. I used CO-STAR to make the social media calendar. After I used it, I saw how it helps me plan my prompt. Then it was not so difficult.

b) Practical Application

I can use prompt engineering in a marketing job. For example, I can use AI to help write emails to customers. I would use the **Role Prompting** technique. I would tell the AI to "Act as a marketing expert." I would also use the **CO-STAR** framework to explain the email's goal and audience. A problem might be that the AI's answer is too general. I will fix this by being very specific in my prompt. I will also check the AI's work and ask it to make the answer better.

c) Limitations & Ethics

AI has two big problems:

1. It can use old or wrong information.
2. It is not a real person and has no feelings.

For ethics, I must be honest. I will not use AI to write fake news or lies to trick customers. I will use AI in a good way. I will always check the information from the AI. I will use it as a tool to help me, not to replace my own work.

d) Goals for the Second Half

I want to learn how to use AI for harder problems. I want to get better at writing prompts so the AI gives me the perfect answer on the first try.